

Crystal Structure

Matter occurs in three states—solid, liquid and gas. Liquid and gas together are called *fluid*. Solid responds elastically to a deforming force. Fluids flow when force is applied and do not return to original configuration when force is removed. But some solids, such as plastic and glass, flow like liquids at high temperatures and behave like solids at low temperatures. The elastic response is not the only criterion.

Solids exhibit crystal structure. The atoms or molecules are arranged in a regular manner over a long distance (long-range order). A substance like glass is more like a liquid than solid from the point of view of atomic arrangement. For a small distance (around a distance up to inter-atomic spacing or so) arrangement of atoms is regular (short-range order). But long-range order is absent.

The solids with ordered arrangement of atoms or molecules are called *crystals* or *crystalline* solids and those without ordered arrangements of atoms or molecules are called *amorphous* solids.

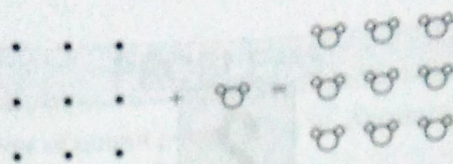
2.1 SPACE LATTICE

A regular periodic configuration of points, particles or objects throughout an area or space is called *lattice* and the points are called *lattice points*. The lattice is a mathematical abstraction. *Space lattice* is a regular, periodic, repeated three dimensional array of points that specifies the position of atoms, molecules or ions in a crystal. Bravais introduced the concept of space lattice therefore sometimes it is called *Bravais* lattice.

A crystal can be reproduced in space by repeating identical structural units. The smallest block is called a *unit cell*. The structural unit can be made up of single atom or several molecules. Every lattice point is associated with a group of atoms called *basis*. The lattice plus its associated basis form the crystal structure. Fig. 2.1 shows two dimensional lattice, basis and structure.

When basis is a single atom, it is called *monoatomic*, when there are two atoms, it is *diatomic* and so on. The smallest cell, which when repeated throughout space will fill all space and produce the lattice, is called *primitive cell*. There are many ways of choosing the primitive axes and primitive cells for a given lattice. Fig. 2.2 shows some possible cells.

A primitive cell is specified by axes a , b and c , called *primitive axes* (Fig. 2.2c). The magnitudes a , b , c are called *lattice parameters*. There is one lattice point per primitive cell. The lattice points are



Space lattice + basis = Crystal structure

Fig. 2.1 Space lattice, basis and crystal structure

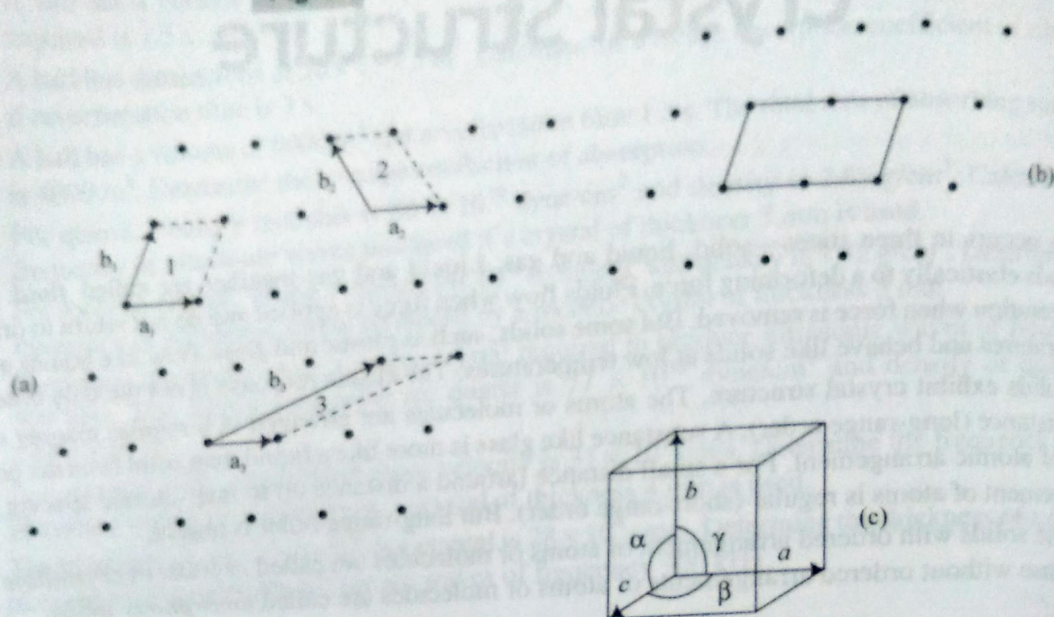


Fig. 2.2 (a) Primitive cells of a two-dimensional lattice (b) non-primitive cell (c) primitive cell in three dimensions

located at the corners of a cell. A primitive cell is a minimum-volume cell. Sometimes it is convenient to specify a cell, larger than minimum-volume cell.

In addition, when the lattice points are on the edges or faces or inside the cell it is called a *non-primitive cell* (Fig. 2.2b). In two dimensions, only five different types of lattices are possible (Fig. 2.3a) but in three dimensions there are fourteen fundamentally different lattices (Fig. 2.3b). These are called Bravais lattices.

2.2 CUBIC STRUCTURES

In cubic systems, $a = b = c$ and $\alpha = \beta = \gamma = 90^\circ$. There are three lattices, single cubic (sc) lattice, the body centered cubic (bcc) lattice and the face centered cubic (fcc) lattice. Fig. 2.4 shows lattice points and atomic arrangements of sc, bcc and fcc.

The primitive cell contains only one lattice point but conventional cubic cell bcc contains two lattice points and fcc contains four lattice points.

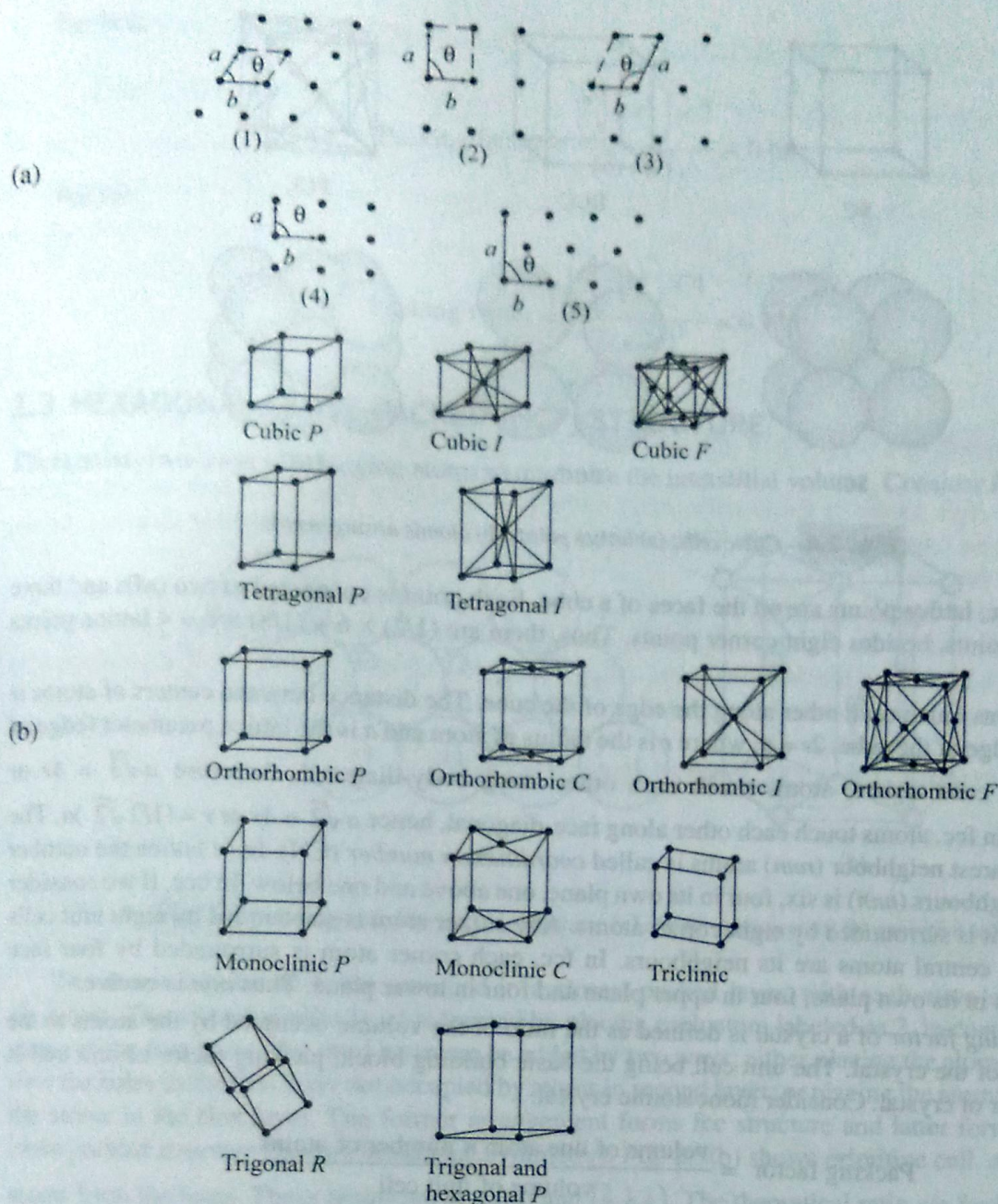
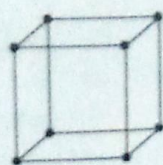


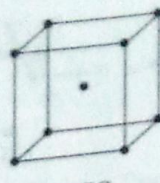
Fig. 2.3 (a) All two-dimensional lattices. 1. $a \neq b$, $\theta \neq 90^\circ$ 2. $a = b$, $\theta = 90^\circ$ 3. $a = b$, $\theta = 120^\circ$ 4. $a \neq b$, $\theta = 90^\circ$ 5. $a \neq b$, $\theta = 90^\circ$ (non-primitive) (b) All three-dimensional lattices. The letters *P*, *I*, *C*, *R* and *F* denote details of cell.

In sc, each corner lattice point is connected to eight different cells, hence it is $1/8$ for each cell and there are eight such points. Thus $(1/8) \times 8 = 1$ lattice point per unit cell. In case of bcc one lattice point is inside the cell and eight lattice points at eight corners. Thus there are $1 + (1/8) \times 8 = 2$ lattice points per

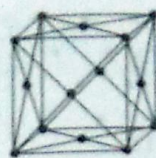
(a)



SC

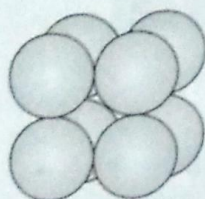


BCC

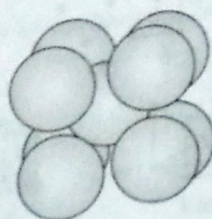


FCC

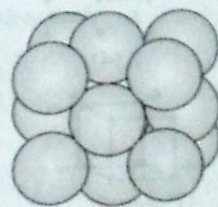
(b)



SC



BCC



FCC

Fig. 2.4 Cubic cells: (a) lattice points (b) atomic arrangements

unit cell. In fcc, lattice points are on the faces of a cube. Each point is connected to two cells and there are six such points, besides eight corner points. Thus, there are $(1/2) \times 6 + (1/8) \times 8 = 4$ lattice points per unit cell.

In sc, atoms touch each other along the edge of the cube. The distance between centers of atoms is equal to the edge of the cube. $2r = a$, where r is the radius of atom and a is the lattice parameter (edge of the cube). In case of bcc, atoms touch each other along body-diagonal, therefore $a\sqrt{3} = 4r$ or $r = (\sqrt{3}/4)a$. In fcc, atoms touch each other along face-diagonal, hence $a\sqrt{2} = 4r$ or $r = (1/2\sqrt{2})a$. The number of nearest neighbour (nnn) atoms is called *coordination number* (CN). In sc lattice the number of nearest neighbours (nnn) is six, four in its own plane, one above and one below. In bcc, if we consider central atom, it is surrounded by eight corner atoms. Any corner atom is surrounded by eight unit cells and the eight central atoms are its neighbours. In fcc, each corner atom is surrounded by four face-centred atoms in its own plane, four in upper plane and four in lower plane. Thus nnn is twelve.

The *packing factor* of a crystal is defined as the ratio of the volume occupied by the atoms to the total volume of the crystal. The unit cell being the basic building block, packing factor of unit cell is packing factor of crystal. Consider monoatomic crystal.

$$\text{Packing factor} = \frac{\text{volume of one atom} \times \text{number of atoms}}{\text{volume of unit cell}}$$

For sc,

$$\text{Packing factor} = \frac{\left(\frac{4}{3}\pi r^3\right) \times 1}{(2r)^3} = 0.524$$

For bcc,

$$\text{Packing factor} = \frac{\left(\frac{4}{3}\pi r^3\right) \times 2}{(4r/\sqrt{3})^3} = 0.68$$

For fcc,

$$\text{Packing factor} = \frac{\left(\frac{4}{3}\pi r^3\right) \times 4}{(2\sqrt{2}r)^3} = 0.734$$